

Transport Poverty Index

A replicable multi-dimensional and multi-modal measure of transport poverty

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Abstract

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Keywords: keyword1, keyword2

1. Introduction

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2. Literature review

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Here are two sample references: [Pereira et al. \(2021\)](#) [INE \(2018\)](#). If citing the other way ([Lovelace et al., 2022](#)).

3. Methods

Add a flowchart

3.1. Each Dimensin's Indicators

3.1.1. Accessibility and so on...

hiuhfjfhjd

3.2. Normalisation

All indicators are normalised to the range $[1, 100]$ using min-max scaling:

Benefit indicators (higher raw value $\rightarrow$ better):

$$X_{norm} = 1 + \frac{X - X_{min}}{X_{max} - X_{min}} \times 99$$

Cost indicators (lower raw value $\rightarrow$ better):

$$X_{norm} = 1 + \frac{X_{max} - X}{X_{max} - X_{min}} \times 99$$

- **Accessibility** metrics (**access_***) are *benefit* indicators.
- **Mobility** cost metrics (**mobility_cost_***), PT waiting times, all **affordability** indicators, and all **safety** indicators are *cost* indicators.

Missing values in accessibility indicators (arising from infeasible PT routes) are imputed with the mean of the respective variable prior to PCA.

3.3. PCA-Based Dimension Scores

For each dimension (Accessibility, Mobility, Safety) and at the global (all-mode) and per-mode levels, a PCA is run using `FactoMineR::PCA()` at parish level.

The **first principal component (PC1)** score is used as the dimension index, after rescaling to $[1, 100]$. PC1 captures the dominant axis of variation among the normalised indicators.

For **Affordability**, PCA is not applied (too few indicators); instead, the normalised composite `h_transp_inc_comp` (housing + transport cost as share of income, modal-share weighted) is used directly as the Affordability Index.

For **Safety per mode**, an equal-weight average of the severity index (per mode) and total vehicles involved (per mode) is used instead of PCA.

For **Mobility - Car** PCA is not applied (too few indicators); instead, we assumed an equal-weight average of the normalized composite `mobility_commuting_avg_tt_car` and `road_length`.

3.4. Composite IMPT Scores

Three aggregation methods are computed for the global IMPT and each per-mode IMPT:

1. **Geometric mean** (equal weights, Figure 1a):

$$IMPT_{geom} = \left(\prod_d (S_d + 1) \right)^{1/n} - 1$$

Partially compensatory: penalizes dimensions with very low values non-linearly. Highlights the most critical situations, where at least one dimension is severely deficient.

2. **Arithmetic mean** (equal weights, Figure 1b):

$$IMPT_{avg} = \frac{1}{n} \sum_d S_d$$

Compensatory: high vulnerability in one dimension can be offset by good performance in others. Suitable for setting aggregate policy targets and communicating to non-specialist audiences.

3. Entropy-weighted geometric mean (Figure 1c):

$$IMPT_{entropy} = \prod_d (S_d + 1)^{w_d} - 1$$

where weights w_d are derived from the **entropy weighting method**:

Weights are determined by the empirical dispersion of the dimensions across parishes: the dimension that varies most among territories receives greater weight. Maximises territorial contrast and is useful for competitive prioritisation of interventions.

$$w_d = \frac{d_d}{\sum_j d_j}, \quad d_d = 1 - e_d, \quad e_d = -k \sum_i p_{id} \ln(p_{id})$$

with $k = 1/\ln(n)$ and p_{id} being the proportion of the dimension score for area i .

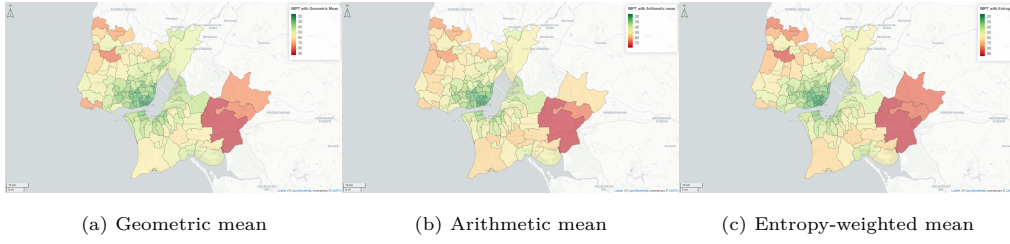


Figure 1: IMPT results for different aggregation methods for all modes (with Navegante)

4. Results

Figure 2 is generated using an R chunk.

5. Discussion

Tables can also be generated using R chunks, as shown in Table 1 example.

```
knitr::kable(head(mtcars)[,1:4])
```

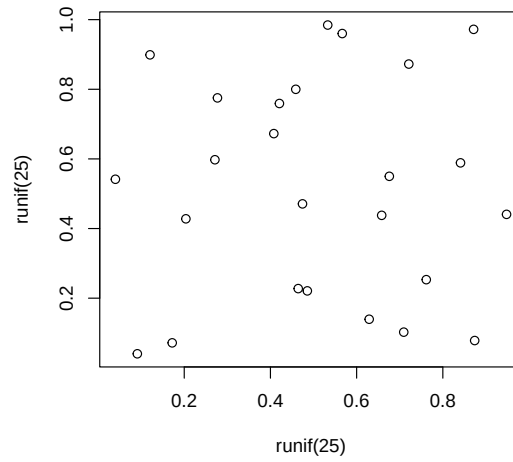


Figure 2: A meaningless scatterplot

Table 1: Caption centered above table

	mpg	cyl	disp	hp
Mazda RX4	21.0	6	160	110
Mazda RX4 Wag	21.0	6	160	110
Datsun 710	22.8	4	108	93
Hornet 4 Drive	21.4	6	258	110
Hornet Sportabout	18.7	8	360	175
Valiant	18.1	6	225	105

6. Conclusions

This is the best paper ever.

References

- INE, I.N.d.E., 2018. Inquérito à mobilidade nas Áreas metropolitanas do porto e de lisboa — imob 2017. URL: <https://www.ine.pt/xurl/pub/349495406>.
- Lovelace, R., Félix, R., Carlino, D., 2022. Jittering: A computationally efficient method

for generating realistic route networks from origin-destination data. Findings doi:[10.32866/001c.33873](https://doi.org/10.32866/001c.33873).

Pereira, R.H.M., Saraiva, M., Herszenhut, D., Braga, C.K.V., Conway, M.W., 2021. r5r: Rapid realistic routing on multimodal transport networks with R5 in R. Findings doi:[10.32866/001c.21262](https://doi.org/10.32866/001c.21262).